



▲ BACTERIAL COLONIES

Colonies of cyanobacteria form rocklike clumps known as stromatolites, which are made of trapped sediment. These ones are in Shark Bay, Western Australia.

ANIMALS AND PLANTS

About 2.2 billion years ago, some single-celled organisms developed special structures that performed different tasks. These organisms are known as protists. In time, some formed colonies that eventually developed into multicelled creatures similar to this jellyfish—the earliest oceanic animals. Life moved onto land about 470 million years ago, leading to the evolution of the first plants, land animals, and fungi.



DAZZLING VARIETY

Over 800 million years, multicelled life has evolved into an amazing diversity of forms. Many have become extinct, including the largest and most spectacular life-forms of all, the dinosaurs. But there are still nearly 2 million different species of animals, plants, and fungi known to science—370,000 of which are species of beetles, including this yellow-spotted fungus beetle.

EVOLUTION AND EXTINCTION

Each generation of a species is slightly different from the previous one. Sometimes this helps them survive, but sometimes it does not. The survivors prosper and breed, while the others die out and disappear. Over time, this mechanism of natural selection leads to the evolution of new species and the extinction of old ones. The ancestors of the modern elephant all evolved in this way, and eventually died out.

